

Probability Problems 2

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Problem 1. (Sampling) Let λ be a positive real number. A random variable X is said to have *exponential distribution with parameter λ* if it has the probability density function

$$f(x) = \begin{cases} \lambda \exp(-\lambda x) & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- Show that $f(x)$ is indeed a probability density function, i.e. that $\int_{-\infty}^{\infty} f(t) dt = 1$.
- Compute the cumulative distribution function F associated with X .
- Compute the inverse of F .
- Write Python code, a function of λ , which samples X at random (n.b. don't use the English word `lambda` for a Python variable of course!). You should use `random.random()` to generate uniform random numbers in the unit interval, but you should use no other built-in methods for generating random numbers or scientific computing packages.
- Check your work by comparing your function to `scipy.stats.expon(scale=1/lam)`. (For example, take a histogram of 1000 numbers chosen according to your distribution and the scipy one).

Problem 2. (Sampling) Let $\alpha > 1$ be a real number. A random variable X is said to be distributed according to a *power law with parameter α* if it has the probability density function

$$f(x) = \begin{cases} (\alpha - 1)x^{-\alpha} & x \geq 1 \\ 0 & x < 1 \end{cases}$$

Repeat all parts of the previous problem for X (the relevant scipy distribution to check against is `scipy.stats.pareto(b=alpha-1)`).

Problem 3. (Sampling) A certain model for the wealth of American households (ignoring debt) suggests that a proportion p of households have no wealth, and the remainder of households have wealth distributed according to a power law with parameter α .

- Let X be the random variable given the wealth of a household chosen at random. Explain why the cumulative distribution function of X is not differentiable.
- Nevertheless, write Python code to sample the wealth of a household (assuming the model), again using only `random.random()`.